



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 10 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Task 10 of the M-Tech Production & Marketing Web Development Internship Program, Batch 2026.

M-Tech Production & Marketing, based in Haripur, KPK, builds modern web apps. After mastering HTML, CSS, and Vanilla JS in Phase 2, we are now moving to Phase 3: Frontend Frameworks & Libraries.

React.js is the #1 library in 2026 for building interactive UIs. Companies use it to build Facebook, Instagram, and most dashboards.

2. Objectives

The main objectives of this task were:

1. To understand what React is and why we use Components.
2. To learn JSX syntax and how React renders to the DOM.
3. To master React Hooks: `useState`, `useEffect`.
4. To learn Props, State Management, and Component Reusability.
5. To learn React Router for multi-page apps.
6. To learn React Context API for global state.
7. To set up a React project using Vite.

3. Work Done

1. Watched Video: "React Course - Beginner to Advanced" by freeCodeCamp - 12 hours.
2. Watched Video: "React JS Crash Course" by Traversy Media.
3. Watched Video: "React Hooks Tutorial" by The Net Ninja.
4. Watched Video: "React Context API" by Web Dev Simplified.
5. Created a React project using Vite: `npm create vite@latest mtech-react-app`
6. Built "M-Tech Task Manager App" in React. Converted Lab 2.4 Vanilla JS app to React Components.
7. Used `useState` for task list, `useEffect` for loading from `localStorage`.
8. Added React Router with 2 pages: Home + About.

4. Challenges Faced

Had to run the react file with first initializing the environment with `npm install` and then to run `npm run dev`

5. Key Learnings

1. What is React: A JavaScript library for building User Interfaces. It uses a "Component" based approach. Everything is a reusable component.
2. JSX: JavaScript + XML. We write HTML inside JS.

```
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const App = () => {
  return <h1 className="text-blue-500">Hello M-Tech</h1>
}
```

3. React Hooks - The

Core:

- useState: To store and update data. `const [count, setCount] = useState(0)`
- useEffect: To run code when component loads. Used for API calls. This replaces a lot of code from Lab 2.5
- 3. Props: How parent components send data to child components. Makes components reusable.
- 4. React Router v6: To make multi-page apps without reloading.

```
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<Routes>
  <Route path="/" element={<Home />} />
  <Route path="/about" element={<About />} />
</Routes>
```

- 5. Context API: To share data like "logged in user" across all components without prop drilling.
- 6. Vite: The new standard for starting React projects. `npm run dev` starts in about 15 second.

6. Conclusion

Phase 3 is where everything from Phase 2 comes together.

React felt hard for the first 2 days, but then Components clicked. The idea of breaking a page into small reusable pieces is powerful.

useState + useEffect are 90% of what you need for most React apps. Async/Await from Lab 2.5 fits perfectly inside useEffect for API calls.

Project: M-Tech Task Manager - React Version

